

Autonomous AI Agent Architectures

Agent Definition An AI Agent is an autonomous system powered by an LLM that can plan, remember context, and execute actions using external tools.

Core Agent Pillars 1. **Planning:** Subgoal decomposition (breaking down complex tasks) and Reflection (refining plans based on environment feedback). 2. **Memory:** Short-term memory (session conversation history) and Long-term memory (reading/writing to vector or relational databases). 3. **Tools:** APIs, search engines, python compilers, and file writers that the model calls to gather info or effect change.

Multi-Agent Systems Multi-agent environments divide complex problems into specialized roles. Agents communicate through message buses or structured interfaces, delegating work dynamically.